

All create table SQL statements

```
CREATE TABLE Station (  
    StationId INT PRIMARY KEY,  
    FIPS VARCHAR(10),  
    Latitude DECIMAL(9,6),  
    Longitude DECIMAL(9,6)  
);
```

```
CREATE TABLE Event (  
    EventId INT AUTO_INCREMENT PRIMARY KEY,  
    StationId INT,  
    Cruise VARCHAR(50),  
    Program VARCHAR(100),  
    Project VARCHAR(100),  
    Agency VARCHAR(100),  
    Source VARCHAR(100),  
    TierLevel VARCHAR(50),  
    FOREIGN KEY (StationId) REFERENCES Station(StationId)  
);
```

```
CREATE TABLE Sample (  
    SampleId INT AUTO_INCREMENT PRIMARY KEY,  
    StationId INT,  
    SampleDate DATE,  
    SampleTime TIME NULL,  
    TotalDepth DECIMAL(10,2) NULL,  
    UpperPycnocline DECIMAL(10,2) NULL,  
    LowerPycnocline DECIMAL(10,2) NULL,  
    Depth DECIMAL(10,2) NULL,  
    Layer VARCHAR(50) NULL,  
    SampleType VARCHAR(50) NULL,  
    SampleReplicateType VARCHAR(10),  
    FOREIGN KEY (StationId) REFERENCES Station(StationId)  
);
```

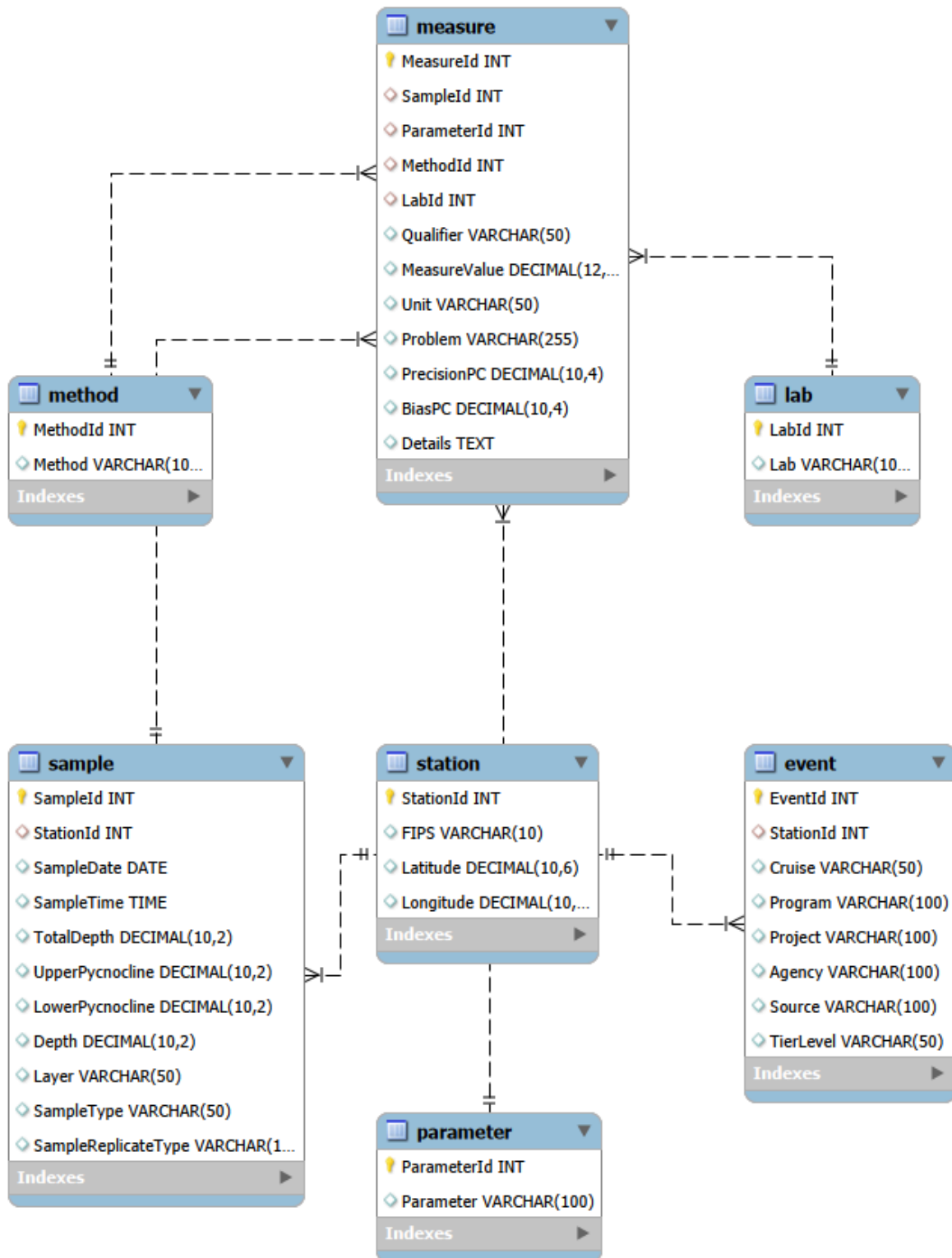
```
CREATE TABLE Parameter (  
    ParameterId INT AUTO_INCREMENT PRIMARY KEY,  
    Parameter VARCHAR(100)  
);
```

```
CREATE TABLE Method (  
    MethodId INT AUTO_INCREMENT PRIMARY KEY,  
    Method VARCHAR(100)  
);
```

```
CREATE TABLE Lab (  
    LabId INT AUTO_INCREMENT PRIMARY KEY,  
    Lab VARCHAR(100)  
);
```

```
CREATE TABLE Measure (  
    MeasureId INT AUTO_INCREMENT PRIMARY KEY,  
    SampleId INT,  
    ParameterId INT,  
    MethodId INT,  
    LabId INT NULL,  
    Qualifier VARCHAR(50) NULL,  
    MeasureValue DECIMAL(12,4) NULL,  
    Unit VARCHAR(50) NULL,  
    Problem VARCHAR(255) NULL,  
    PrecisionPC DECIMAL(10,4) NULL,  
    BiasPC DECIMAL(10,4) NULL,  
    Details TEXT NULL,  
    FOREIGN KEY (SampleId) REFERENCES Sample(SampleId),  
    FOREIGN KEY (ParameterId) REFERENCES Parameter(ParameterId),  
    FOREIGN KEY (MethodId) REFERENCES Method(MethodId),  
    FOREIGN KEY (LabId) REFERENCES Lab(LabId)  
);
```

A screenshot of the ERD



Screenshots of the first 5 rows of data from TWO tables, showing all columns.

271
272
273
274
275

[illegible]

278
279
280
281

[illegible]

All SQL queries for each of the four questions.

Q1: What is the average CHLA (chlorophyll a) concentration for each month? How does it compare to the overall average CHLA value for the entire dataset? (Hint: Use the column MeasureValue for CHLA concentration.)

```
SELECT
    MONTH(s.SampleDate) AS Month,
    AVG(m.MeasureValue) AS Avg_CHLA,
    (SELECT AVG(m2.MeasureValue)
     FROM Measure m2
     JOIN Parameter p2 ON m2.ParameterId = p2.ParameterId
     WHERE p2.Parameter = 'CHLA') AS Overall_Avg_CHLA
FROM Measure m
JOIN Sample s ON m.SampleId = s.SampleId
JOIN Parameter p ON m.ParameterId = p.ParameterId
WHERE p.Parameter = 'CHLA'
GROUP BY MONTH(s.SampleDate)
ORDER BY Month;
```

Q2: What is the maximum and minimum CHLA concentration at each station, and when (date and time) were the measurements taken? (Hint: Use the column MeasureValue for CHLA concentration.)

```
SELECT
    st.StationId,
    'MAX' AS Type,
    m.MeasureValue,
    s.SampleDate,
    s.SampleTime
FROM Measure m
JOIN Sample s ON m.SampleId = s.SampleId
JOIN Parameter p ON m.ParameterId = p.ParameterId
JOIN Station st ON s.StationId = st.StationId
WHERE p.Parameter = 'CHLA'
AND m.MeasureValue = (
    SELECT MAX(m2.MeasureValue)
    FROM Measure m2
    JOIN Sample s2 ON m2.SampleId = s2.SampleId
    JOIN Parameter p2 ON m2.ParameterId = p2.ParameterId
    WHERE p2.Parameter = 'CHLA'
    AND s2.StationId = st.StationId
)
```

UNION

```
SELECT
    st.StationId,
    'MIN' AS Type,
    m.MeasureValue,
    s.SampleDate,
    s.SampleTime
FROM Measure m
JOIN Sample s ON m.SampleId = s.SampleId
JOIN Parameter p ON m.ParameterId = p.ParameterId
JOIN Station st ON s.StationId = st.StationId
WHERE p.Parameter = 'CHLA'
AND m.MeasureValue = (
    SELECT MIN(m2.MeasureValue)
    FROM Measure m2
    JOIN Sample s2 ON m2.SampleId = s2.SampleId
    JOIN Parameter p2 ON m2.ParameterId = p2.ParameterId
    WHERE p2.Parameter = 'CHLA'
    AND s2.StationId = st.StationId
)
```

)
ORDER BY StationId, Type;

Q3: How many samples were taken for each SampleReplicateType (S1, FS1, etc.) across all stations?

```
SELECT
    SampleReplicateType,
    COUNT(*) AS Sample_Count
FROM Sample
GROUP BY SampleReplicateType;
```


Q4: Are there any stations where the CHLA value consistently does not exceed a certain threshold (18.0 ug/L) throughout the year? If so, which stations, and during which months does this occur?

```
SELECT
    st.StationId,
    MONTH(s.SampleDate) AS Month
FROM Measure m
JOIN Sample s ON m.SampleId = s.SampleId
JOIN Parameter p ON m.ParameterId = p.ParameterId
JOIN Station st ON s.StationId = st.StationId
WHERE p.Parameter = 'CHLA'
GROUP BY st.StationId, MONTH(s.SampleDate)
HAVING MAX(m.MeasureValue) <= 18.0
ORDER BY st.StationId, Month;
```